CLASS X · PRACTICE EXAMINATION

BIOLOGY — MODEL QUESTION PAPER 1

Final I pattern · Section A (40) + Section B (60)

Code: BCSE/22-3/MODEL-1 Time: 2 hours 30 minutes

Total: 100 marks Pages: see footer

Instructions. (1) Section A is compulsory: Question 1 carries 40 marks. (2) In Section B attempt **any SIX** questions (10 marks each). (3) Write the answers in the spaces given; use a pencil for diagrams and graphs. (4) Circle only **ONE** letter in the multiple-choice questions. (5) The figures are drawn for practice and are not to scale.

SECTION A [40 MARKS]

ANSWER **ALL** QUESTIONS

Question 1 **[25]**

a) **Directions: For each question there are four alternatives A, B, C and D. Choose the correct alternative and circle it neatly. If more than one letter is circled, NO score will be awarded.** **[25]**

i. Which structure is present in both plant cells and animal cells?

A Cell wall
B Chloroplast
C Mitochondrion
D Large central vacuole

ii. *Helicobacter pylori* infection of the stomach can be diagnosed by

A an electrocardiogram.
B a spirometer test.
C a blood sugar test.
D a urea breath test.

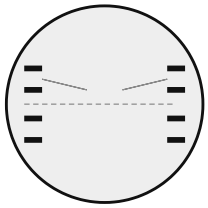
iii. A plant cell placed in a hypertonic solution

A becomes turgid.
B does not change.
C bursts.
D becomes plasmolysed.

iv. Root hair cells absorb mineral ions from soil that has a lower concentration of the ions than the cell. This happens by

A active transport.
B diffusion.
C osmosis.
D transpiration.

v. The figure shows a cell during division. Which phase is shown?



A Prophase
B Metaphase
C Telophase
D Anaphase

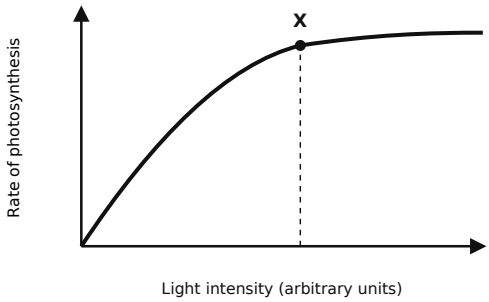
vi. Which products of the light reaction are used in the Calvin cycle?

A Oxygen and glucose
B ATP and NADPH
C Carbon dioxide and water
D Starch and ATP

vii. The light reaction of photosynthesis takes place in the

A grana.
B stroma.
C mitochondria.
D cytoplasm.

viii. The graph shows the effect of light intensity on the rate of photosynthesis. Beyond point X the rate does not rise. Which factor is most likely limiting?



A Light intensity
B Oxygen concentration
C Carbon dioxide concentration
D Amount of soil

ix. A difference between xylem vessels and phloem sieve tubes is that xylem vessels

A are made of dead cells.
B carry food.
C have companion cells.
D carry substances in both directions.

x. Sugars are loaded into the phloem by

A osmosis.
B active transport.
C diffusion.
D transpiration pull.

xi. Digestion of starch begins in the mouth because saliva contains

A salivary amylase.
B pepsin.
C lipase.
D trypsin.

xii. The function of bile in digestion is to

A emulsify fats.
B digest proteins.
C digest starch.
D kill bacteria in the stomach.

xiii. Villi in the small intestine are adapted for absorption because they

A kill bacteria.
B secrete bile.
C increase the surface area.
D produce enzymes for protein digestion only.

xiv. A person of blood group B has

A A antigen and anti-B antibody.
B B antigen and anti-A antibody.
C A and B antigens and no antibody.
D no antigen and both antibodies.

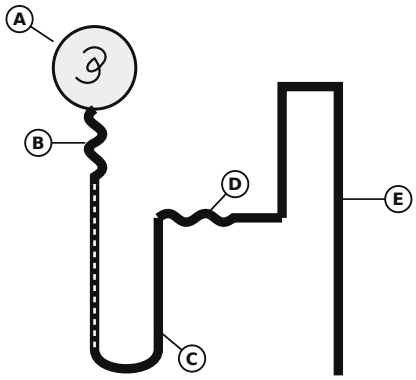
xv. A patient of blood group AB can safely receive blood from

A AB only.
B A and B only.
C O only.
D A, B, AB and O.

xvi. An invasive species is a species that

A is protected by law.
B is not native, spreads quickly and harms native species.
C is rare in the wild.
D is found nowhere else in the world.

xvii. In the nephron shown, all the glucose in the filtrate is reabsorbed in the part labelled



A A
B B
C C
D E

xviii. The hormone that increases reabsorption of water in the collecting duct is

A insulin.
B adrenaline.
C ADH.
D gibberellin.

xix. Which of the following is a reflex action?

A Blinking when dust enters the eye
B Writing a letter
C Riding a bicycle
D Reading a book

xx. Gametes are produced by

A meiosis.
B mitosis.
C binary fission.
D budding.

xxi. A diploid cell has 8 chromosomes. After meiosis each gamete has

A 2
B 4
C 8
D 16

xxii. Tall (TT) pea plants are crossed with dwarf (tt) plants and the F_1 plants are self-pollinated. The F_2 ratio of tall : dwarf is

A 1 : 1
B 3 : 1
C 1 : 3
D 2 : 1

xxiii. Colour blindness is more common in males than in females because

A the gene is on the Y chromosome.
B the gene is dominant.
C males have two X chromosomes.
D males have only one X chromosome.

xxiv. Bacteria become resistant to an antibiotic because

A they decide to survive.
B the antibiotic makes them resistant.
C resistant bacteria survive and reproduce.
D they learn to avoid it.

xxv. Which is an example of ecosystem diversity?

A Different breeds of cattle
B Genes of a tiger population
C Many varieties of rice
D Coral reef, desert and forest

b) Fill in the blanks with suitable word/s. **[5]**

i.	The loss of water vapour from the leaves through the stomata is called _____.	
ii.	The end product of protein digestion is _____.	
iii.	The cell fragments that help blood to clot are _____.	
iv.	A random change in allele frequency in a small population is called genetic _____.	
v.	The movement of sugars from the leaves to other parts of a plant is called _____.	

c) Write **TRUE** or **FALSE** for each statement in the 'Answer' column. **[5]**

i.	Osmosis does not need energy from ATP.		
ii.	The dark reaction can continue for a long time without the products of the light reaction.		
iii.	A spinal reflex is controlled mainly by the spinal cord.		
iv.	Mitosis produces genetically different cells.		
v.	A large, randomly mating population with no mutation, migration or selection keeps constant allele frequencies.		

d) Match each item in **Column A** with the most appropriate item in **Column B**. Write the correct letter in the 'Answer' column. **[5]**

Answer	Column A	Column B	
i.	i. Controls reabsorption of water in the collecting duct	a. Stroma	
ii.	ii. Use of microbes to clean up an oil spill	b. Bioremediation	
iii.	iii. Exchange of alleles between populations	c. ADH	
iv.	iv. Site of the Calvin cycle	d. Haemoglobin	
v.	v. Carries oxygen in red blood cells	e. Gene flow	
		f. Thylakoid	
		g. Platelets	

SECTION B [60 MARKS]

ATTEMPT **ANY SIX** QUESTIONS

Question 2

a) A student set up potato strips of equal length (6.0 cm) in sugar solutions of different concentrations. The final lengths after one hour are given. **[3]**

Sugar solution (%)	**0**	**10**	**20**	**30**	**40**
Final length (cm)	6.6	6.3	6.0	5.7	5.4

Plot a line graph of final length against concentration of sugar solution.

b) Which solution is isotonic to the potato cells? Give a reason. **[1]**

c) Explain why the strip in 0 % sugar solution became longer. **[2]**

d) State ONE way to make the results more reliable. **[1]**

e) A gardener adds too much fertiliser (salt) around the roots of vegetables and the plants wilt. Explain. **[2]**

f) How is facilitated diffusion different from simple diffusion? **[1]**

Question 3

a) The graph shows the effect of light intensity on the rate of photosynthesis of a water plant. (See the graph in Q1(a)(viii).) **[1]**

(i) Name the independent variable and the dependent variable.

(ii) Explain the shape of the graph. **[2]**

(iii) Suggest how a greenhouse farmer could increase the rate beyond X. **[2]**

b) Write the word equation (or chemical equation) for photosynthesis. **[1]**

c) Give TWO differences between the light reaction and the dark reaction. **[2]**

d) Why is photosynthesis important to animals? **[1]**

e) State the role of chlorophyll. **[1]**

Question 4

a) A ring of bark is removed completely around the stem of a woody plant and the plant is observed for a few months. **[2]**

(i) Draw a labelled diagram to show the expected result.

(ii) Explain the result. **[2]**

b) Name the process by which food is moved in a plant. Give ONE example each of a source and a sink. **[2]**

c) State THREE differences between xylem and phloem. **[3]**

d) Name the plant hormone used (i) to stop potatoes from sprouting in storage and (ii) to ripen fruits early. **[1]**

Question 5

a) Complete the table. **[2]**

Food	**Enzyme**	**Organ**	**End product**
Starch	(i)	Mouth	(ii)
Protein	Pepsin	(iii)	Peptides
Fat	Lipase	Small intestine	(iv)

b) State the role of hydrochloric acid in the stomach. **[1]**

c) (i) What is *Helicobacter pylori*? (ii) Why is it a concern in Bhutan? (iii) How is it treated? **[3]**

d) Give TWO ways to prevent constipation. **[1]**

e) How is the small intestine adapted for absorption? **[2]**

f) Why is bile not an enzyme but still important in digestion? **[1]**

Question 6

a) A doctor mixed blood samples P, Q, R and S with anti-A and anti-B antibodies. **[2]**

Sample	**Anti-A**	**Anti-B**
P	clumps	clumps
Q	clumps	no clumping
R	no clumping	no clumping
S	no clumping	clumps

Identify the blood group of P, Q, R and S.

b) Which sample is the universal donor? Give a reason. **[1]**

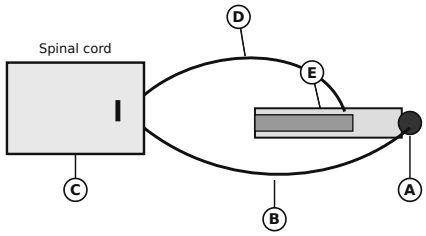
c) Explain why a person of group A cannot receive blood of group B. **[2]**

d) State ONE function of platelets and ONE adaptation of red blood cells. **[2]**

e) Give TWO ways in which HIV spreads and ONE way to reduce stigma against people living with HIV. **[3]**

Question 7

a) The figure shows the pathway of a reflex action. **[2]**
(i) Name the parts A to E.



(ii) Write the sequence of the pathway when the hand touches a hot object. **[2]**

b) Why is a reflex action faster than a voluntary action? **[1]**

c) Relate the following parts of a computer to the nervous system: (i) keyboard, (ii) CPU, (iii) monitor, (iv) hard disk. **[2]**

d) Why do some athletes use anabolic steroids? Give TWO harmful effects. **[3]**

Question 8

a) A pure tall pea plant (TT) is crossed with a pure dwarf plant (tt). Use Punnett squares to show the F_1 and the F_2 generations. State the phenotypic ratio of the F_2. **[3]**

b) A woman who is a carrier of colour blindness (X^CX^c) marries a colour-blind man (X^cY). (i) Draw a Punnett square. (ii) What percentage of their daughters are colour-blind? **[3]**

c) A diploid cell of an animal has 12 chromosomes. How many chromosomes will be present in (i) each daughter cell after mitosis, (ii) each gamete after meiosis? **[1]**

d) Give TWO differences between mitosis and meiosis. **[2]**

e) Explain how meiosis produces genetic variation. **[1]**

Question 9

a) In an industrial area the percentage of dark peppered moths changed as shown. **[1]**

Year	**1850**	**1870**	**1890**	**1910**
Dark moths (%)	5	25	60	85

(i) State the trend.

(ii) Explain the trend using natural selection. **[3]**

b) Two populations of squirrels were separated by a newly formed river. Name the type of isolation and explain how it can lead to a new species. **[3]**

c) In a population, the frequency of the recessive allele (q) is 0.4. Calculate the frequency of the dominant allele (p) and of heterozygous individuals. **[2]**

d) Name ONE invasive plant in Bhutan and state ONE problem it causes. **[1]**

Rough Work

CLASS X · PRACTICE EXAMINATION

BIOLOGY — MODEL QUESTION PAPER 2

Final I pattern · Section A (40) + Section B (60)

Code: BCSE/22-3/MODEL-2 Time: 2 hours 30 minutes

Total: 100 marks Pages: see footer

Instructions. (1) Section A is compulsory: Question 1 carries 40 marks. (2) In Section B attempt **any SIX** questions (10 marks each). (3) Write the answers in the spaces given; use a pencil for diagrams and graphs. (4) Circle only **ONE** letter in the multiple-choice questions. (5) The figures are drawn for practice and are not to scale.

SECTION A [40 MARKS]

ANSWER **ALL** QUESTIONS

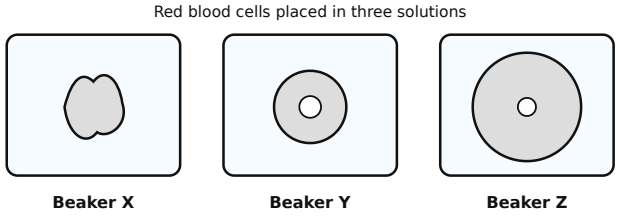
Question 1 **[25]**

a) **Directions: For each question there are four alternatives A, B, C and D. Choose the correct alternative and circle it neatly. If more than one letter is circled, NO score will be awarded.** **[25]**

i. Prokaryotic cells differ from eukaryotic cells because prokaryotic cells

A have a nucleus.
B have chloroplasts.
C are larger.
D lack membrane-bound organelles.

ii. Red blood cells were placed in three solutions as shown. Which beaker contains a solution that is hypertonic to the cells?

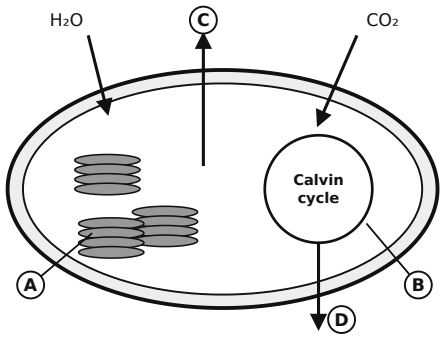


A Beaker X
B Beaker Y
C Beaker Z
D None of them

iii. Which process needs ATP?

A Osmosis
B Active transport
C Simple diffusion
D Facilitated diffusion

iv. The figure shows a chloroplast. In which part does the light reaction take place?



A A
B B
C C
D D

v. The raw materials of the light reaction are

A water and light.
B carbon dioxide and ATP.
C glucose and oxygen.
D NADPH and ATP.

vi. Which hormone can be sprayed to stop potatoes from sprouting in storage?

A Gibberellin
B Ethylene
C NAA (auxin)
D Cytokinin

vii. Which hormone is used to ripen fruits early?

A 2,4-D
B Ethylene
C Abscisic acid
D Gibberellin

viii. Which statement about the phloem is correct?

A Its cells are dead.
B It carries water only.
C Its walls contain lignin.
D It carries sugars in both directions.

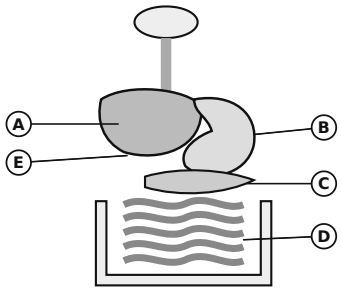
ix. After a ring of bark is removed from a stem, the stem swells above the ring because

A water collects there.
B sugars cannot move down past the ring.
C minerals collect there.
D the xylem is blocked.

x. Pancreatic juice contains enzymes that digest

A only starch.
B only protein.
C starch, protein and fat.
D only fat.

xi. In the figure, which letter shows the organ that makes bile?



A A
B B
C C
D D

xii. The final product of carbohydrate digestion that is absorbed in the small intestine is

A maltose.
B fatty acids.
C amino acids.
D glucose.

xiii. A person with gallstones that block the bile duct has difficulty digesting

A starch.
B proteins.
C vitamins.
D fats.

xiv. Which blood group is the universal donor?

A AB
B A
C B
D O

xv. The plasma of a person of blood group O contains

A anti-A only.
B anti-B only.
C anti-A and anti-B.
D no antibodies.

xvi. Phagocytosis in blood is carried out by

A white blood cells.
B red blood cells.
C platelets.
D plasma.

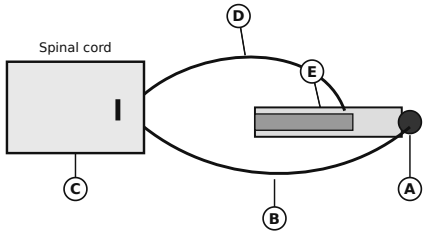
xvii. Which substance is completely reabsorbed in the kidneys of a healthy person?

A Water
B Urea
C Salts
D Glucose

xviii. In the human kidney, filtration of blood takes place in the

A loop of Henle.
B glomerulus and Bowman's capsule.
C distal convoluted tubule.
D collecting duct.

xix. The figure shows a reflex arc. Which letter represents the motor neuron?

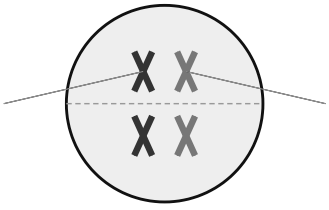


A A
B B
C C
D D

xx. Crossing over occurs during

A anaphase II.
B metaphase I.
C prophase I.
D telophase I.

xxi. The figure shows a cell in division. The phase shown is



A metaphase I.
B metaphase II.
C metaphase of mitosis.
D prophase I.

xxii. A heterozygous tall plant (Tt) is crossed with a dwarf plant (tt). The percentage of tall offspring is

A 0 %
B 25 %
C 50 %
D 100 %

xxiii. A woman who is a carrier of haemophilia marries a normal man. The chance that a son is haemophilic is

A 0 %
B 25 %
C 50 %
D 100 %

xxiv. Which is an example of geographic isolation?

A Two plant species flower in different seasons.
B A river splits a squirrel population.
C Birds with different courtship calls.
D Species living in different habitats of the same area.

xxv. Bhutan is part of the biodiversity hotspot called the

A Western Ghats.
B Sundaland.
C Eastern Himalaya.
D Madagascar.

b) Fill in the blanks with suitable word/s. **[5]**

i.	The enzyme that digests protein in the stomach is _____.	
ii.	The plant hormone _____ closes stomata during drought.	
iii.	A trait that shows only when two copies of the recessive allele are present is called a _____ trait.	
iv.	The use of microorganisms to clean up pollution is called _____.	
v.	The structure in the prokaryotic cell where the DNA is found is the _____.	

c) Write **TRUE** or **FALSE** for each statement in the 'Answer' column. **[5]**

i.	Mitochondria are absent in prokaryotic cells.		
ii.	In active transport substances move from a region of low concentration to high concentration.		
iii.	The dark reaction takes place in the grana.		
iv.	Gene flow increases the differences between populations.		
v.	Lantana camara is an invasive plant in Bhutan.		

d) Match each item in **Column A** with the most appropriate item in **Column B**. Write the correct letter in the 'Answer' column. **[5]**

Answer	Column A	Column B	
i.	i. Breaks fats into tiny droplets	a. Prophase I	
ii.	ii. Site of filtration in the nephron	b. Bile	
iii.	iii. Crossing over of chromosomes	c. Glomerulus	
iv.	iv. A small group starts a new population	d. Ethylene	
v.	v. Hormone that ripens fruits	e. Founder effect	
		f. Stroma	
		g. Loop of Henle	

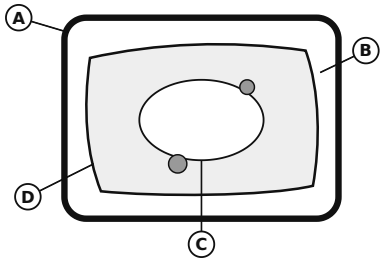
SECTION B [60 MARKS]

ATTEMPT **ANY SIX** QUESTIONS

Question 2

a) The figure shows a plant cell that was placed in a strong salt solution. **[1]**

(i) Name the parts A and C.



(ii) Name the process that caused the change and state what fills the space B. **[1]**

(iii) What would happen to this cell if it was then placed in pure water? **[1]**

b) Give TWO differences between active transport and facilitated diffusion. **[2]**

c) Dry raisins placed in water become swollen after a few hours. Explain. **[2]**

d) Draw a labelled diagram of a prokaryotic cell. Label the part that is not found in eukaryotic cells. **[3]**

Question 3

a) A student measured the number of oxygen bubbles released by pondweed in 5 minutes at different carbon dioxide concentrations. **[3]**

$NaHCO_3$ (g/100 ml)	**0**	**0.2**	**0.4**	**0.6**	**0.8**
Bubbles in 5 min	6	20	34	40	40

Plot a graph of the results.

b) Name the independent and dependent variables. **[1]**

c) State a conclusion from the results. **[1]**

d) Suggest TWO ways a farmer can improve the yield of tomatoes in a greenhouse using the knowledge of photosynthesis. **[2]**

e) What is the role of water in the light reaction? **[1]**

f) Explain why photosynthesis stops at night even though the dark reaction does not need light. **[2]**

Question 4

a) A ring of bark was removed from the stem of a woody plant. Draw a diagram to show the result after some months and explain. **[3]**

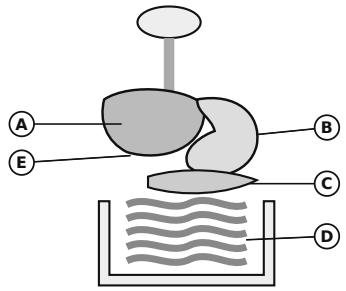
b) Why are leaves called a source and roots a sink? **[2]**

c) State THREE differences between xylem and phloem. **[3]**

d) Give TWO uses of auxin in farming. **[2]**

Question 5

a) Study the figure of the human digestive system. Name the parts A, B, C and D. **[2]**



b) Where is bile stored and what is its function? **[2]**

c) Explain what would happen to digestion if the pancreatic duct was blocked. **[2]**

d) State TWO habits that improve digestion. **[2]**

e) Why is chewing important? **[2]**

Question 6

a) State ONE function each of red blood cells, white blood cells and platelets. **[3]**

b) A man of blood group AB needs a blood transfusion. Which groups can he receive and why? **[2]**

c) The table shows the concentration (g/100 ml) of some substances in the glomerular filtrate and in urine. **[3]**

	Glucose	**Urea**	**Protein**
Filtrate	0.10	0.03	0
Urine	0	2.00	0

Explain the differences for glucose and urea.

d) What would happen if a person did not make enough ADH? **[2]**

Question 7

a) (i) Trace the pathway of a reflex when a person steps on a sharp pin. **[2]**

(ii) Why is the brain not needed for the first response? **[1]**

b) Give TWO differences between a reflex action and a voluntary action. **[2]**

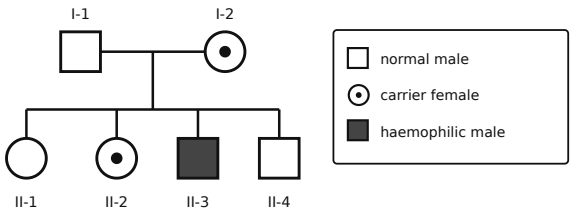
c) How is the human nervous system similar to a computer? Give TWO points. **[2]**

d) State THREE harmful effects of anabolic steroids. **[3]**

Question 8

a) The figure shows a family in which haemophilia occurs. **[1]**

(i) Write the genotype of I-2 and II-3.



(ii) From whom did II-3 inherit the haemophilia allele? Explain. **[2]**

b) Two heterozygous tall pea plants (Tt) are crossed. Draw a Punnett square and give the genotypic and phenotypic ratios. **[3]**

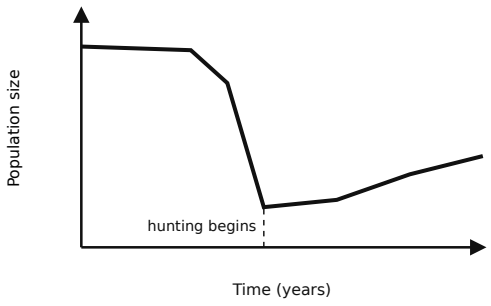
c) Explain how meiosis keeps the chromosome number the same from one generation to the next. **[2]**

d) A cell has 14 chromosomes. How many are present in each cell after (i) mitosis, (ii) meiosis? **[1]**

e) Why are the daughter cells formed by mitosis identical? **[1]**

Question 9

a) The graph shows the population of an animal before and after hunting began. **[1]**
(i) Name the effect that reduced the genetic variation of the survivors.



(ii) Explain the effect on the genetic diversity of the population. **[2]**

b) A river divides a bird population into two. Name the type of isolation and explain how it can lead to speciation. **[3]**

c) What is an invasive species? Give ONE example from Bhutan and ONE effect on the ecosystem. **[3]**

d) State ONE way Bhutan conserves its biodiversity. **[1]**

Rough Work

ANSWER KEYS

Model Paper 1 and Model Paper 2

ANSWER KEY — MODEL PAPER 1

BCSE/22-3/MODEL-1 · for self-checking

Section A — Q1(a) Multiple choice (1 mark each)

No.	Answer	Reason
i.	**C**	Mitochondria are found in all eukaryotic cells; cell wall, chloroplast and large vacuole are plant features.
ii.	**D**	The urea breath (or stool antigen) test detects H. pylori.
iii.	**D**	Water leaves by osmosis, the membrane pulls away from the wall: plasmolysis.
iv.	**A**	Movement against the concentration gradient needs ATP: active transport.
v.	**D**	Sister chromatids have separated and are moving to opposite poles: anaphase.
vi.	**B**	ATP gives energy and NADPH gives hydrogen to reduce CO_2 to glucose.
vii.	**A**	Chlorophyll is in the thylakoid membranes of the grana.
viii.	**C**	Light is no longer limiting; CO_2 (or temperature) now limits the rate.
ix.	**A**	Xylem vessels are dead, hollow, lignified tubes carrying water upward only.
x.	**B**	Loading of sucrose into the phloem needs energy from the companion cells.
xi.	**A**	Salivary amylase changes starch to maltose.
xii.	**A**	Bile breaks fat into tiny droplets, increasing the area for lipase.
xiii.	**C**	Villi and microvilli give a very large absorbing surface.
xiv.	**B**	Group B: B antigen on the red cells, anti-A in the plasma.
xv.	**D**	AB plasma has no antibodies, so AB is the universal recipient.
xvi.	**B**	Invasive species are non-native organisms that spread rapidly and damage ecosystems.
xvii.	**B**	B is the proximal convoluted tubule, which reabsorbs all glucose and amino acids (A = Bowman's capsule, C = loop of Henle, D = DCT, E = collecting duct).
xviii.	**C**	Antidiuretic hormone (ADH) makes the collecting duct more permeable to water.
xix.	**A**	Blinking is rapid, automatic and involuntary.
xx.	**A**	Meiosis halves the chromosome number to make gametes.
xxi.	**B**	Haploid number = 8 ÷ 2 = 4.

xxii.	**B**	Tt × Tt gives TT, Tt, Tt, tt = 3 tall : 1 dwarf.
xxiii.	**D**	A male needs only one recessive allele on his single X chromosome.
xxiv.	**C**	Natural selection: resistant mutants survive and pass on the allele.
xxv.	**D**	Different ecosystems in a region = ecosystem diversity.

Q1(b) Fill in the blanks · Q1(c) True/False · Q1(d) Matching (1 mark each)

	(b)	(c)	(d)
i.	**transpiration**	**TRUE**	**c** — ADH
ii.	**amino acids**	**FALSE**	**b** — Bioremediation
iii.	**platelets**	**TRUE**	**e** — Gene flow
iv.	**drift**	**FALSE**	**a** — Stroma
v.	**translocation**	**TRUE**	**d** — Haemoglobin

Section B (10 marks per question; any six)

Question 2

Part	Marks	Model answer and marking points
a)	3	Axes: x = concentration of sugar solution (%), y = final length (cm) with units (1); correct plotting of the five points (1); ruled line / smooth line through the points and a title (1). The line slopes downwards from 6.6 cm to 5.4 cm.
b)	1	**20 %**; the length did not change (6.0 cm), so there is no net movement of water.
c)	2	Pure water is **hypotonic** to the cell sap (1). Water enters the cells by **osmosis** through the selectively permeable membrane, so the cells become turgid and the strip lengthens (1).
d)	1	Repeat the experiment with several strips in each solution and calculate the **average** (or use strips of the same size, same temperature and time).
e)	2	Excess fertiliser makes the soil solution **hypertonic** to the root cells (1). Water leaves the root cells by osmosis, cells lose turgor and the plant wilts (1).
f)	1	Facilitated diffusion needs **carrier or channel proteins**; simple diffusion passes directly through the lipid bilayer. (Neither uses ATP.)

Question 3

Part	Marks	Model answer and marking points
a)	1	Independent: **light intensity**; dependent: **rate of photosynthesis** (e.g. bubbles of oxygen per minute).
	2	Rate rises at first because more light means more ATP and NADPH made in the light reaction (1). At X another factor (CO_2 or temperature) becomes limiting, so the rate levels off (1).
	2	Raise the **CO_2** concentration (burn fuel/CO_2 enrichment) (1) and keep the temperature near the optimum (35–40 °C) / provide water (1).
b)	1	Carbon dioxide + water → (light, chlorophyll) glucose + oxygen. ($6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$)

c)	2	Light reaction: in the **grana**, needs **light and water**, produces ATP, NADPH and O_2. Dark reaction: in the **stroma**, needs CO_2, ATP and NADPH, produces glucose. (any two, 1 mark each)
d)	1	It provides **food** (glucose) at the start of food chains and **oxygen** for respiration.
e)	1	Chlorophyll **absorbs light energy** and converts it into chemical energy.

Question 4

Part	Marks	Model answer and marking points
a)	2	Stem with a bulge/swelling **above** the ring, no swelling below; labels: ring (gap) where bark is removed, swelling, bark/phloem, xylem, leaves green. (1 mark drawing, 1 mark labels)
	2	The ring removes the **phloem** (1). Sugars made in the leaves cannot move down past the gap and accumulate above it, causing swelling; the xylem is intact so water still rises (1).
b)	2	**Translocation** (1). Source: mature leaf; sink: root / fruit / growing bud / seed (1).
c)	3	Xylem: carries water and minerals; dead cells; upward only; no ATP needed. Phloem: carries food (sucrose); living cells; up and down; needs ATP. (any three, 1 mark each)
d)	1	(i) **Auxin (NAA)** (ii) **Ethylene** (0.5 mark each).

Question 5

Part	Marks	Model answer and marking points
a)	2	(i) salivary amylase; (ii) maltose; (iii) stomach; (iv) fatty acids and glycerol. (0.5 mark each)
b)	1	Kills bacteria in food and provides the acid medium for pepsin; activates pepsinogen to pepsin. (any one)
c)	3	(i) A spiral bacterium living in the stomach lining that causes gastritis and ulcers (1). (ii) High rate of infection and gastric cancer in Bhutan (1). (iii) Triple therapy (two antibiotics + acid-reducing drug) for 14 days (1).
d)	1	Eat fibre (vegetables, fruits, whole grain); drink plenty of water; regular exercise (any two, 0.5 each).
e)	2	Very long and coiled; wall folded into villi and microvilli giving a large surface area (1); thin one-cell lining and rich capillary/lacteal supply (1).
f)	1	Bile has no enzymes but it **emulsifies fats** so lipase can act faster.

Question 6

Part	Marks	Model answer and marking points
a)	2	P = **AB**, Q = **A**, R = **O**, S = **B** (0.5 mark each).
b)	1	**R (group O)** — it has no A or B antigens, so no clumping occurs in any recipient.
c)	2	Group B red cells carry B antigen (1). The recipient's plasma has anti-B antibody, which attaches to the antigen and makes the red cells clump (agglutination) (1).
d)	2	Platelets help the blood to clot (1). RBCs have no nucleus / biconcave shape / contain haemoglobin to carry more oxygen (1).
e)	3	Spreads by unprotected sex, sharing needles or shaving blades, infected blood, mother to child (any two, 1 mark each). Stigma: awareness education / counselling and equal treatment (1).

Question 7

Part	Marks	Model answer and marking points
a)	2	A = receptor (skin), B = sensory neuron, C = spinal cord (relay neuron), D = motor neuron, E = effector (muscle). (5 correct = 2, 3–4 correct = 1)
	2	Receptor → sensory neuron → interneuron (spinal cord) → motor neuron → effector (muscle) → hand pulled away.
b)	1	The signal passes through the spinal cord only and not through the brain, so the pathway is shorter.
c)	2	(i) receptors / sense organs; (ii) brain and spinal cord; (iii) effectors (muscles, glands); (iv) memory in the brain. (0.5 mark each)
d)	3	To build muscle and body mass quickly (1). Harmful effects: infertility, heart/liver damage, high blood pressure, aggression, depression, mood swings (any two, 1 mark each).

Question 8

Part	Marks	Model answer and marking points
a)	3	F_1: all Tt, all tall (1). F_2 (Tt × Tt): TT, Tt, Tt, tt (1). Phenotypic ratio **3 tall : 1 dwarf** (1).
b)	3	(i) Gametes X^C, X^c × X^c, Y gives X^CX^c, X^cX^c, X^CY, X^cY (2). (ii) Daughters: X^CX^c (carrier) and X^cX^c (colour-blind) → **50 %** (1).
c)	1	(i) **12** (ii) **6** (0.5 mark each).
d)	2	Mitosis: 1 division, 2 identical diploid cells, in body cells. Meiosis: 2 divisions, 4 different haploid cells, in germ cells, with crossing over. (any two, 1 mark each)
e)	1	Crossing over in prophase I and random (independent) assortment of homologous chromosomes give different combinations of alleles in each gamete.

Question 9

Part	Marks	Model answer and marking points
a)	1	The percentage of dark moths **increased** steadily (5 % to 85 %).
	3	Soot darkened the tree trunks (1). Dark moths were better camouflaged and were eaten less by birds, so more survived and reproduced (1). They passed the dark allele to offspring, so its frequency increased (1).
b)	3	**Geographic isolation** (1). No gene flow; mutations and different selection pressures make the populations different (1) until they can no longer interbreed = speciation (1).
c)	2	$p = 1 - 0.4 =$ **0.6** (1); $2pq = 2 \times 0.6 \times 0.4 =$ **0.48** (1).
d)	1	Lantana camara / crofton weed (Ageratina) / polka dot plant / Mikania — crowds out native plants and reduces pasture and crop yield.

ANSWER KEY — MODEL PAPER 2

BCSE/22-3/MODEL-2 · for self-checking

Section A — Q1(a) Multiple choice (1 mark each)

No.	Answer	Reason
i.	**D**	Prokaryotes have no nucleus or membrane-bound organelles (DNA lies in the nucleoid).
ii.	**A**	In X the cells have shrunk, so water left by osmosis: the solution is hypertonic.
iii.	**B**	Active transport moves substances against the gradient using ATP.
iv.	**A**	A = grana (thylakoids), B = stroma, C = oxygen released, D = glucose.
v.	**A**	Light energy splits water in the light reaction.
vi.	**C**	NAA prevents early sprouting.
vii.	**B**	Ethylene promotes fruit ripening.
viii.	**D**	Phloem carries sucrose up and down to wherever it is needed.
ix.	**B**	The phloem is cut, so sugars from the leaves accumulate above the gap.
x.	**C**	Pancreatic amylase, trypsin and lipase digest all three.
xi.	**A**	A = liver, B = stomach, C = pancreas, D = small intestine.
xii.	**D**	Maltose is changed to glucose by maltase.
xiii.	**D**	Without bile, fats are not emulsified.
xiv.	**D**	Group O red cells have no A or B antigens.
xv.	**C**	Group O has both antibodies and no antigens.
xvi.	**A**	White blood cells engulf pathogens.
xvii.	**D**	All the glucose is reabsorbed in the PCT; urea is not.
xviii.	**B**	High pressure in the glomerulus forces the filtrate into the capsule.
xix.	**D**	A = receptor, B = sensory neuron, C = spinal cord, D = motor neuron, E = effector.
xx.	**C**	Homologous chromosomes pair and exchange segments in prophase I.
xxi.	**A**	Homologous pairs (bivalents) line up at the equator: metaphase I.
xxii.	**C**	Tt × tt gives 1 Tt : 1 tt = 50 % tall.
xxiii.	**C**	Half of the sons receive the X chromosome with the haemophilia allele.
xxiv.	**B**	A physical barrier (river) separates the populations.
xxv.	**C**	The Eastern Himalaya hotspot includes Bhutan, Nepal and north-east India.

Q1(b) Fill in the blanks · Q1(c) True/False · Q1(d) Matching (1 mark each)

	(b)	(c)	(d)

i.	**pepsin**	**TRUE**	**b** — Bile
ii.	**abscisic acid**	**TRUE**	**c** — Glomerulus
iii.	**recessive**	**FALSE**	**a** — Prophase I
iv.	**bioremediation**	**FALSE**	**e** — Founder effect
v.	**nucleoid**	**TRUE**	**d** — Ethylene

Section B (10 marks per question; any six)

Question 2

Part	Marks	Model answer and marking points
a)	1	A = **cell wall**; C = **vacuole** (0.5 mark each).
	1	**Osmosis** (plasmolysis); space B is filled with the outside (salt) solution.
	1	Water enters by osmosis, the vacuole swells and pushes the membrane against the wall; the cell becomes turgid again (deplasmolysis).
b)	2	Active transport: against the gradient, uses ATP, carrier (pump) proteins. Facilitated diffusion: down the gradient, no ATP, uses channel/carrier proteins. (any two, 1 mark each)
c)	2	Raisins contain a concentrated sugar solution, so water is hypotonic to them (1). Water enters by osmosis through the semipermeable skin and the raisins swell (1).
d)	3	Rod/oval cell with cell wall, cell membrane, cytoplasm, ribosomes, plasmid, flagellum (2); **nucleoid** (DNA region with no nuclear membrane) labelled clearly (1).

Question 3

Part	Marks	Model answer and marking points
a)	3	x-axis: sodium hydrogen carbonate (g/100 ml); y-axis: number of bubbles in 5 min, with scale and units (1); points plotted correctly (1); smooth curve rising then flat, with title (1).
b)	1	Independent: **CO_2 concentration ($NaHCO_3$)**; dependent: **number of bubbles (rate of photosynthesis)**.
c)	1	As CO_2 concentration rises the rate increases up to 0.6 g/100 ml, after which it stays constant (another factor limits).
d)	2	Keep the temperature near the optimum (30–35 °C); give enough light (transparent roof / lamps); raise CO_2; water regularly. (any two)
e)	1	Water is **split (photolysis)** to give hydrogen for NADPH and oxygen as a by-product.
f)	2	At night the light reaction stops, so no ATP and NADPH are made (1). The dark reaction depends on these products and soon runs out of them (1).

Question 4

Part	Marks	Model answer and marking points
a)	3	Swelling above the ring, no swelling below (1, labelled). Phloem was removed so sugars from the leaves cannot move down (1). Xylem is intact, water still moves up (1).

b)	2	Leaves make sugar by photosynthesis in excess of their needs and export it (source) (1). Roots use or store the sugar (sink) (1).
c)	3	Xylem: water and minerals, dead cells, upward only. Phloem: sugars, living cells, both directions, needs ATP. (any three)
d)	2	NAA stops early fruit drop / sprouting of potatoes; helps cuttings to make roots; 2,4-D kills broad-leaved weeds. (any two)

Question 5

Part	Marks	Model answer and marking points
a)	2	A = liver; B = stomach; C = pancreas; D = small intestine (0.5 mark each).
b)	2	Stored in the **gall bladder** (1); **emulsifies fats** and makes the food alkaline (1).
c)	2	Pancreatic amylase, trypsin and lipase cannot reach the small intestine (1), so digestion of starch, proteins and fats is reduced (1).
d)	2	Chew food thoroughly; eat at regular times; exercise regularly; drink water; eat fibre (any two).
e)	2	Chewing breaks food into small pieces, increasing the surface area for enzymes (1) and mixes it with saliva containing amylase (1).

Question 6

Part	Marks	Model answer and marking points
a)	3	RBC: carry oxygen (haemoglobin); WBC: defend the body / phagocytosis; platelets: help clotting. (1 mark each)
b)	2	A, B, AB and O (1). AB has no anti-A or anti-B in the plasma, so no clumping occurs (1).
c)	3	Glucose is completely reabsorbed in the PCT, so none is in urine (1). Urea is not reabsorbed while water is, so its concentration is much higher in urine (1). Some urea is also secreted (1).
d)	2	Less water is reabsorbed in the collecting duct (1), so a large volume of dilute urine is passed and the person becomes dehydrated (1).

Question 7

Part	Marks	Model answer and marking points
a)	2	Receptor (skin) → sensory neuron → interneuron in spinal cord → motor neuron → effector (leg muscle) → leg is withdrawn.
	1	The spinal cord processes the signal so the response is quick.
b)	2	Reflex: involuntary, fast, spinal cord, no thinking, e.g. blinking. Voluntary: controlled by the brain, slower, e.g. writing. (any two)
c)	2	Receptors/sensors give input; brain/CPU processes; effectors/monitor give output; electrical signals; memory/storage. (any two)
d)	3	Infertility, irregular periods, liver and heart damage, high blood pressure, mood swings, aggression, depression (any three).

Question 8

Part	Marks	Model answer and marking points
a)	1	I-2 = X^HX^h (carrier); II-3 = X^hY (0.5 mark each).
	2	From his **mother (I-2)** (1). A son receives his X chromosome from his mother and the Y chromosome from his father (1).
b)	3	Gametes T, t × T, t gives TT, Tt, Tt, tt (1). Genotypic ratio 1 TT : 2 Tt : 1 tt (1). Phenotypic ratio 3 tall : 1 dwarf (1).
c)	2	Meiosis halves the number in gametes (1); fertilisation restores the diploid number in the zygote (1).
d)	1	(i) 14 (ii) 7 (0.5 mark each).
e)	1	The DNA is copied exactly and each chromatid is shared equally between the two cells; no crossing over.

Question 9

Part	Marks	Model answer and marking points
a)	1	**Bottleneck effect** (genetic drift).
	2	Only a few individuals survive, so many alleles are lost (1). The population becomes less variable and is less able to adapt to disease or change (1).
b)	3	**Geographic isolation** (1). No gene flow; mutation, natural selection and drift act differently on each population (1) until they cannot interbreed (1).
c)	3	A non-native species that spreads quickly and harms native species (1). Example: Lantana camara / crofton weed / polka dot plant (1). Effect: crowds out native plants, reduces biodiversity or crop yield (1).
d)	1	60 % forest cover for all time in the Constitution; protected areas and biological corridors; hunting ban; community forests (any one).

Marking guide: in Section B each bracketed point is worth 1 mark unless shown; accept any correct answer in different words. For graphs award marks for axes with units, correct plotting and the line/curve. For diagrams award marks for the drawing and for correct labels.